

The Raw Ham's Lite Logger

Complete Guide - Install, Setup & Operating

A small field logger and self-spotter for POTA, WWFF, SOTA, GMA, WWBOTA and LLOTA with radio control (Yaesu, Icom, Xiegu and more), offline callbook, worked-before checks, and sending to QRZ or your own logger.

For Windows 10 / 11, Android and Linux · Setup takes about 10 minutes

Quick start

It's free from the download page, **rawhamradio.github.io/rawham-lite-logger**. Grab the download for your computer or tablet, then follow its box below. **No Python or other downloads needed** on Windows or Android.

Windows 10 / 11 - about 2 minutes

1. Download **Raw_Ham_Lite_Logger_Setup.exe** and double-click it.

2. If a blue box says **"Windows protected your PC"**, click **More info** → **Run anyway**. (It says that for any new program that isn't from a big company.)

3. Click **Next**, then **Install**, then **Finish**. The logger starts, and its icon is on your desktop and in the Start menu.

4. Do the **first-time setup** (Step 2) - the Settings window opens by itself.

Android tablet or phone - about 2 minutes

1. On the tablet, open the download page and tap **RawHamLiteLogger.apk**. (Or copy the file over from a PC with a USB cable, Google Drive, or email it to yourself.)

2. Tap the file. If Android asks, allow **Install unknown apps** for your Files app or browser, then tap **Install**.

3. Open **Raw Ham Lite Logger** and tap **Settings** - see Part 3.

Linux (Ubuntu, Mint, Raspberry Pi ...) - about 5 minutes

1. Download **Raw_Ham_Lite_Logger_Linux.zip** and unzip it. Open the **Raw Ham Lite Logger Linux** folder, right-click inside it → **Open in Terminal**.

2. Type `bash install-linux.sh` and press Enter. Answer **Y** to its questions.

3. Start it from your app menu or the desktop icon - see Part 4.

Please don't share your own files

Send friends the download page link. Never send **rawhams_lite_data.json** or **rawhams_lite.db** - they hold **your** settings, log and passwords. Each person's logger makes its own.

Part 1 - Installing and setting up

Before you start

You will need:

1. A Windows 10 or 11 laptop, an Android tablet or phone, or a Linux computer.
2. An internet connection for the first-time setup (afterwards it works with no signal).
3. About **300 MB** of free space for the program and its offline lists.
4. Optional: a **QRZ.com subscription** (name lookups and the QRZ logbook), **your home logging program** (ACLog, Log4OM, HRD, DXKeeper, Wavelog ...) and a **USB/CAT cable** for radio control.

Step 1 - Install it on Windows

1. Download **Raw_Ham_Lite_Logger_Setup.exe** from the download page, **rawhamradio.github.io/rawham-lite-logger**.
2. Double-click it. If Windows shows **"Windows protected your PC"**, click **More info** → **Run anyway**.
3. Leave **Put an icon on the desktop** ticked and click **Next** → **Install** → **Finish**. No administrator password is needed and **Python is not needed**.
4. From now on, start it with the **desktop icon** or from the **Start menu**. The Start menu also has this guide.

Using Linux, or the Python file on Windows? See the Appendix at the end. On a tablet, see Part 3.

Step 2 - First-time setup (do this at home, online)

The first time it opens, the logger shows **Settings**. Everything is saved, so you only do this once. Later, open it with the **Settings** button (top right) or by clicking your call.

1. **My station** tab: type your **call** and **home grid**. Keep **US** as the POTA country (add others like **US, CA** if you travel). Tick **Dark mode** if you like a dark screen (easier on the eyes at night).
2. **QRZ** tab (optional): your **Logbook API key** (QRZ → Logbook → Settings) for Send to QRZ and worked-before, and your QRZ **username and password** for name lookups. Click **Test QRZ**. Both need a QRZ subscription.
3. **Log of choice** tab (optional): pick the logging program you use at home - see Step 4.
4. **Offline data** tab: click **Update everything now**. This downloads the park and summit lists, your worked-before history, and (first time) the offline US/Canada callbook. The callbook takes several minutes.

5. Leave **When the logger opens online, update automatically** ticked. Then every time you open the logger at home it refreshes worked-before from QRZ and your log - so the list is current before you leave the QTH.

> **Settings tabs at a glance**
> **My station** - call, grid, countries · **Spotting** - spot comment, re-spot after each QSO · **Radio** - CAT port and speed · **QRZ** - API key and login · **Log of choice** - where Send to LOC goes · **WSJT-X** - FT8/FT4 listener · **Offline data** - updates, imports, export folder.

Step 3 - Connect your radio (CAT control)

The logger follows your radio's frequency and mode, and can tune it. Any radio can still log and spot without this.

1. Connect the radio's USB (or CAT) cable and turn the radio on.
2. Open **Settings** → **Radio** and pick your radio in the **Radio** list. The boxes underneath change to match it.
3. Pick the **CAT port** (COM port) and set the speed to match the radio's CAT menu. Tick **Connect to the radio when the logger starts** if you like, then click Done.
4. Click **Connect radio** at the top. The status should say **Connected:** and your model. Turn the VFO - the frequency in the logger follows it.

Radio	Settings	Notes
Yaesu FT-710, FT-991A, FT-891, FTDX10, FTDX101	**Enhanced** COM port, 38400	FT-710: tick **DTR/RTS off**. FT-991A: speed is menu 031.
Yaesu FT-817, FT-818, FT-857, FT-897	CAT cable's COM port, 4800 (or the CAT RATE menu)	FT8/FT4 use **DIG** mode - set the radio's DIG MODE menu to **USER-U**.
Icom IC-705, IC-7300, IC-7300 MK2, IC-7100, IC-7610, IC-9700	The radio's USB COM port, 19200	CI-V address fills in for you, and is found automatically if it's different. Turn CI-V USB Echo Back OFF if you can. FT8 switches the radio to USB-D (data).
Xiegu G90, X6100, X6200, X5105, X108G	The CI-V cable's COM port, 19200	Uses the Icom language (CI-V). G90 address 88, X6100/X6200 A4, X5105 70.
Any other radio (Windows)	**Flrig** (port 12345) or **Hamlib rigctld** (port 4532)	Start Flrig or rigctld first, set up for your radio. The logger then talks to it on this PC. Covers Kenwood, Elecraft, FlexRadio and hundreds more.

To move the radio from the logger, type a frequency (like **14.074** or **14074**) in **Freq kHz** and press Enter, or pick a mode.

Step 4 - Where your contacts go: QRZ and your log of choice

There are two send buttons at the bottom of the window: **Send to QRZ** and **Send to** your **log of choice** (the button shows its name, like **Send to ACLog**). Use either or both. The log of choice is handy if you don't have a paid QRZ subscription.

Pick it in **Settings** → **Log of choice**, fill in the boxes that appear, and click **Test connection**:

Logger	Set up in that program	Worked-before over Wi-Fi?
N3FJP ACLog (any N3FJP logger)	Settings → Application Program Interface → TCP API Enabled. Port 1100. 127.0.0.1 if on the same PC, or the PC's address from a tablet.	**Yes**
Wavelog / Cloudlog	Site address, an API key with write access, and the station location number.	**Yes**
DXLab DXKeeper	Configuration → Defaults → Network Service on. Port 52001.	No - import ADIF
Log4OM 2	Add a UDP inbound connection for WSJT-X messages. Port 2237 (use another if WSJT-X also sends here).	No - import ADIF
HRD Logbook	QSO Forwarding, receive UDP on 2333.	No - import ADIF
Other loggers	Anything that takes WSJT-X "Logged ADIF" or raw ADIF over UDP (GridTracker, MacLoggerDX ...).	No - import ADIF

Tick **Also send each QSO the moment I log it** to send as you go. Each contact is marked **Q** (on QRZ) or **L** (in your log of choice) and is never sent twice. N1MM Logger+ and Nexus can't receive contacts over the network - use Export ADIF and import the file.

> **Worked-before - update it before you leave home**
> Worked-before comes from your **QRZ logbook** (with an API key), your **ACLog log file** on this PC, **ACLog over Wi-Fi**, and **Wavelog/Cloudlog**. It updates when the logger opens online, or click **Settings** → **Offline data** → **Update worked-before**. For DXKeeper, Log4OM, HRD and others, export an ADIF and use **Import worked-before from an ADIF file**. The logger never changes your home log file.

Part 2 - Using the logger

The screen has four areas, top to bottom: **Station & radio** (your call, grid and the **Connect radio** and **Settings** buttons), **Activation** (your parks, frequency, mode and SPOT ME), **Log a contact**, and the **log table**. Buttons along the bottom export and send your log.

Your park or summit

Your grid follows the park. The park and summit lists include each one's location, so when you enter a reference the logger uses **that park's grid square** as your grid in the ADIF file, QRZ and your log of choice - the line under your reference shows it ("my grid EM15fl"). If a park's location isn't known, your home grid is used. Turn this off in Settings → My station.

1. Type it in **Reference(s)**. For a POTA park in your home country **just the number** works: **1234** means US-1234. Other programs use their full reference: **KFF-1234** (WWFF), **W5O/OU-001** (SOTA), **B/G-0001** (WWBOTA).

2. As you type, a **list drops down** with matching parks and their names. Use the **arrow keys** and **Enter**, or click one. Typing part of a name (like **thunder**) works too, and so does the **Find...** button.
3. **Several references** (an n-fer, or a park that is also WWFF): separate them with commas, like **1234, KFF-1234**. The line underneath shows each one with its name.
4. **GMA and LLOTA** references need the program in front: **GMA:DL/EW-017**, **LLOTA:US-0001**.

Suggestions and names come from lists saved on the laptop, so they work with no signal. They refresh themselves whenever you're online and they're over 30 days old. GMA, WWBOTA and LLOTA references log and spot fine but don't show names.

Spotting yourself - SPOT ME

Program	What SPOT ME does
POTA, WWBOTA	Posts your spot directly.
GMA	Sends the spot through the GMA cluster.
WWFF	Tries the WWFF cluster; if that's closed, opens the WWFF spot page with the details copied (paste with Ctrl+V).
SOTA, LLOTA	Need your own login, so the spot page opens with the details already copied - paste with Ctrl+V.

The spot comment is in **Settings** → **Spotting** - leave it blank for an automatic one. Spots are limited to one a minute. **Re-spot after each QSO** only uses POTA, WWBOTA and GMA, so no browser windows pop up mid-pileup (turn it on in Settings → Spotting). To practice, use park **K-TEST** and mark it QRT on pota.app afterwards.

Logging contacts

The top line is **Call Sent Rcvd P2P Comment Log**, with **Name**, **State** and **Country** underneath. It works the same on Windows, Linux and Android.

1. Type the **Call**. Press **Tab** to move along: Call → Sent → Rcvd → P2P → Comment → back to Call (**Enter** or the LOG button logs, so Tab skips it). Each box's text is **highlighted** when you arrive, so just type over it. **Shift+Tab** goes back.
2. Press **Enter** in any box (or click **Log**) to log. **Esc** clears the form.
3. **Sent / Rcvd** fill with 59 (phone), 599 (CW) or -10 (FT8/FT4) when you change mode.
4. **Comment** is optional - a note for this contact. It's saved with the QSO, shown in the log, and goes into the ADIF file and to QRZ / your log of choice.
5. **Name**, **State** and **Country** fill in by themselves (Country from the call's prefix, even offline), so Tab skips them. Click them to type over - your typing is kept.
6. Logging the same station on the same band and mode twice in a UTC day asks before saving.
7. To **fix** a contact, double-click it (or right-click → **Edit QSO**). Every field can be changed; nothing is saved until you click **Save**. To remove one, press **Delete** or right-click it.
8. The big number counts today's contacts (UTC) and turns green at the goal: **10** for POTA, **44** for WWFF, **4** for SOTA. It restarts at 0000 UTC, the same way POTA counts each UTC day.

Who am I talking to? - the lookup line

While you type a call, the line beside **Name** shows where the station is, where that came from, and your history with them:

Tag	Comes from
[FCC]	The offline US callbook (Download offline callbook).
[ISED]	The offline Canada callbook.
[QRZ]	A QRZ lookup - live when online, then saved so it works offline next time.
[your log]	A name you logged before (from QRZ, your home log or this logger).

After that it shows **Worked 3x before, last 2025-06-12 20m SSB** and either **NEW on 20m SSB** or **already worked on 20m SSB** - or **First QSO with this station!** If you left a comment on an earlier contact with them (here, on QRZ, in ACLog or Wavelog), your most recent note shows too, like **your note (2025-06-12): "QRP from US-1234"**.

Park-to-park (P2P)

1. Put the other station's park in **P2P**. Numbers, names and the drop-down list work here exactly like your own park.

2. If they're at **several parks**, type a comma after the first and add the next. Each one is listed underneath, numbered, with its name.
3. In your log it's one row. When exported it becomes **one record per park**, a second apart - that's how POTA gives credit for each park.

Finishing - export and upload

1. Click **Export ADIF...**. One file per reference is saved in **Documents\Raw Ham Logger ADIF** and the folder opens with the file highlighted.
2. Drag each file onto the program's upload page.

Program	File name	Upload at
POTA	KJ5GVQ@US-1234-20261018.adi	pota.app → My Logs
WWFF	KJ5GVQ@KFF-1234 20261018.adi	WWFF logsearch / your national coordinator
SOTA	KJ5GVQ@W5O_OU-001-20261018.adi	sotadata.org.uk
GMA, WWBOTA, LLOTA	CALL@REF-DATE.adi	Their websites

A **/** can't be used in Windows file names, so SOTA summits and portable calls like KJ5GVQ/P use _ in the file name. The call inside the log is unchanged. Change the folder in Settings → Offline data → **Choose ADIF export folder**.

3. Click **Send to QRZ** and/or **Send to** your log of choice (see Step 4). The number on each button is how many contacts still need sending; sent ones are marked and never sent twice.
4. Click **New activation** to clear the list for next time. It saves a copy of the list to a file first, so a wrong click never loses a log. Your worked-before history is kept.

Past activations

1. Click **Past activations...** at the bottom. Your exported activations are listed, newest first.
2. Double-click one to open it. Double-click a contact (or click **Edit**) to fix it, or **Delete** it, then click **Save file**. The first time, the original is kept as a .bak copy.
3. Use **Send to QRZ** or **Send to** your log of choice to push that activation. QRZ replaces any copy it already has; some loggers would get duplicates if they already have them.
4. Click **Continue** to bring that activation back on screen, just like you never left: keep logging, Share it, send it. Handy if you hit New activation too soon. Whatever was on screen gets saved to a file first. On the tablet: Past activations, tap the file, then **Continue**.

FT8 / FT4 from WSJT-X or Nexus

Every FT8/FT4 contact you log in WSJT-X - or in **Nexus** - goes straight into the activation with your park attached. Turn it on in **Settings** → **WSJT-X** (port 2237). In WSJT-X: Settings → Reporting →

UDP Server 127.0.0.1 port 2237. If GridTracker or JTAlert also listen, use the shared (multicast) address they use - 224.0.0.73, 224.0.0.1 and 239.255.0.1 are heard automatically. In Nexus, turn on its WSJT-X UDP output (127.0.0.1:2237). For safety the logger only takes contacts from WSJT-X on **this PC**; if WSJT-X runs on another computer, tick **Also accept contacts from other computers** in the WSJT-X tab.

Settings - every tab

Tab	What's in it
My station	Callsign, home grid, POTA countries for the park lists, use the park's grid square, **dark mode**, **Special event call** and **Operator**.
Spotting	Spot comment (blank = automatic), Re-spot me after each QSO, allow DX cluster spots when not activating, and the DX cluster address.
Radio	Radio type, CAT port and speed (or CI-V address, or Flrig/rigctld address), DTR/RTS off, connect at startup.
QRZ	Logbook API key (Send to QRZ, worked-before) and username/password (name lookups and photos). Test QRZ. **LoTW station location** (picked from TQSL's list).
Log of choice	Which logger the Send to LOC button uses, its address/port or Wavelog details, the name on the button, send-each-QSO, and Test connection.
WSJT-X	Listen for FT8/FT4 contacts, port, multicast address, and whether other computers may send contacts.
Offline data	What's downloaded and when; update automatically; Update everything now; Update worked-before; Update park & summit lists; Download offline callbook; Choose ACLog log file; Import worked-before from ADIF; Choose ADIF export folder.

The **Settings** menu at the top also has **Find a park or summit** and **Update everything now**.

What's new - Windows, Linux and Android

The station bar now has On/Off buttons for these (the Raw Roster is on Windows and Linux only). Each one does nothing at all while it's Off, so the logger stays light on slow laptops.

Button	What it does
Activating: On / Off	On = at a park or summit (as before). Off = logging from home or the field - see below.
Spots: On / Off	Live POTA, SOTA, WWFF and DX spots (needs internet). Click one to tune the radio and fill in Call and P2P.
Raw Roster: On / Off	Everyone WSJT-X is decoding, as a list - any WSJT-X mode (FT8, FT4, Q65, JT65 ...). Double-click a station and WSJT-X calls them.
Photo: On / Off	The station's QRZ photo pops up when you Tab out of Call (needs internet and the QRZ login). Drag it wherever you like - it remembers the spot. It closes when you log the contact.

On a tablet

Spots, Activating, Photo, Share, special event calls, Country, the automatic backup and the Activated! alert all work on Android too. When Activating is Off, the tablet has a **Use GPS** button that fills in your grid (no internet needed). The red **CLR** key on the ham keyboard clears the contact. Share opens the tablet's share sheet so you can post straight to Facebook, Messages or email. The backup files are in Download → Raw Ham Logger → Backup.

Not at a park - Activating: Off

1. Click **Activating: On** to turn it Off. The park box goes away and **My grid** takes its place. If WSJT-X is running, your grid comes from WSJT-X; otherwise type it (blank = your home grid).
2. The **P2P** box becomes **Their park** - for a park you hunted. It counts for hunter credit, and it is never marked P2P.
3. **Export ADIF** saves a daily log like **KJ5GVQ-20261025.adi** with no park in it, ready for QRZ, LoTW or your logger. Turn Activating back On and your last park comes back.

Special event calls (Route 66 On The Air and others)

1. Put the event call (like **W6J**) in **Settings** → **My station** → **Special event call**. The station bar turns orange: **W6J (op KJ5GVQ)**.
2. Every contact is logged as the event call with you as the operator, marked **E** in the list.
3. **Send to QRZ, Send to LOC and Send to LoTW switch off**, so the event log can't go to your own logbooks by mistake. **Export ADIF** makes the event's own file (like **W6J-20261025.adi**) - send that to the event.

4. Using the event call in WSJT-X? Those contacts are recognized as the event call by themselves.
5. Clear the box when the event is over.

SPOT ME when not activating goes to a DX cluster instead of the OTA sites, and only if you tick **Allow SPOT ME to the DX cluster** in Settings → Spotting. Self-spotting on DX clusters is frowned on and is against the rules in contests, so use it rarely, for special events only - it allows one spot every 15 minutes. FT8 already shows up on PSK Reporter and CW on the Reverse Beacon Network by themselves.

Raw Roster (WSJT-X)

Turn on **Listen to WSJT-X** in Settings → WSJT-X, then click **Raw Roster: On**. It works in every WSJT-X mode with contacts - FT8, FT4, FT2, Q65, JT65, JT9, MSK144 and FST4. Each station shows who they're **calling** (CQ, CQ POTA, or the station they're working), their **grid**, **country**, signal and age, tagged **NEW**, **B4**, **DUPE** or **P2P**. Click a heading to sort by it (default newest first). Anyone **calling you** turns red and stays on top. Double-click a station and WSJT-X calls them. On a slow laptop you can close WSJT-X's waterfall (Wide Graph) and work from the Raw Roster.

About / Credits in the Settings menu shows your version number - handy when reporting a problem.

Share - a map picture for social media

Click **Share** at the bottom. It draws a map of the stations you worked (with lines from your park) and saves the picture, copies a ready-made post to the clipboard ("KJ5GVQ at US-1234 - 23 QSOs ..."), and opens the folder. Paste the words into Facebook or your club's group and attach the picture. Stations are placed by their grid, park, callbook, state or country; any with no known location are left off.

Spot comment and QRT

The **Spot comment** box sits right beside SPOT ME - type something like *QRP 5W* or *CW only* and it goes with your spots (blank = an automatic one). When you're shutting down, press the red **QRT** button: it sends a last spot for each of your parks, so hunters stop calling, and automatic re-spots stop. QRT uses the same comment box - type *thanks all, 73* and it goes out as *QRT - thanks all, 73* (QRT is added so the spot sites mark you off the air). Leave it blank and it sends *QRT - thanks all!*. Pressing SPOT ME again puts you back on the air.

Spot them - spotting the station you're working

SPOT ME (the big green button up top) spots **you**. **Spot them** (next to LOG) spots the station in the **Call** box - handy when you're hunting and want to help an activator. It uses their park in the P2P / Their park box and your frequency and mode, and it's sent as you. Type a comment first if you like (e.g. *tnx 59 in OK*). It always asks before sending.

Their park	Where the spot goes
POTA	Straight onto pota.app
WWFF, GMA, WWBOTA	Their spot clusters / site, sent as you

Their park	Where the spot goes
SOTA, LLOTA	Their spot page opens with the spot copied - paste it (those sites need you signed in)
No park (DX)	The DX cluster in Settings → Spotting. Spotting *other* stations is what the cluster is for, so this is always fine.

Send to LoTW

1. Install ARRL's **TQSL** and set up your certificate and a Station Location in it (one time).
2. In **Settings** → **QRZ**, pick your **LoTW station location** from the list (it's read from TQSL).
3. Click **Send to LoTW**. TQSL signs and uploads; your certificate never leaves TQSL. Sent contacts are marked **W**.

Also new

1. **Updates** - your version number sits at the right end of the top row. Grey means you're up to date. **Red** means a newer one is out: click it (or tap it on Android) to download and install. Your log and settings are kept. PC also has Settings → Check for updates.
2. **Activated!** - at 10 POTA (4 SOTA, 44 WWFF) contacts the counter flashes green and beeps. The counter also shows your rate, like **23/hr**.
3. **Midnight UTC** - a reminder at 23:50Z, and after 0000Z an offer to export yesterday's log and start the new day's count from 0.
4. **Automatic backup** - every contact is written to a daily backup file the moment you log it (the **backup** folder next to your settings).
5. **Operator** - handing the logger to a friend? Put their call in Settings → My station → Operator, and clear it when you take it back.
6. **Find my radio...** in Settings → Radio lists the COM ports with their full names, and the **Disconnect** button (red) lets go of the radio.

Part 3 - The Android app

The same logger runs on an Android phone or tablet (Android 8 or newer). It works offline in the field and controls Yaesu radios through a USB-OTG cable.

Installing

1. On the tablet, download **RawHamLiteLogger.apk** (from the link KJ5GVQ shares, or copy the file over).
2. Tap it. If Android asks, allow **Install unknown apps** for your browser or Files app, then tap **Install**.
3. To update later, install the new APK the same way - your log and settings are kept.

Setting up

Appearance in card 1 picks light, dark, or follows the tablet's own dark-mode setting. Tap **Settings**. It is one page with numbered cards, the same as the Windows tabs: **1 My station**, **2 Spotting**, **3 Radio**, **4 QRZ**, **5 Log of choice** and **6 Offline data**. At home on Wi-Fi, tap **Update everything now** in card 6 before you leave.

Logging on the tablet

The form is Call, Sent, Rcvd, then P2P, then Comment with the **LOG** button, then Name, State and Country. The red **CLR** key on the ham keyboard clears the contact (like Esc on a PC). For capitals in a comment, tap ■ once for one capital, or twice for Caps Lock. **abc** switches to the tablet's own keyboard; put it away (down arrow or Back) and the ham keyboard comes back.

1. Tap a logging box and the **ham keyboard** appears: four short rows with the **numbers along the top**, QWERTY below, and ***/***, **-**, **,** and **.** keys for portable calls, parks and reports. The keys are made as big as possible while the whole logging form stays visible above them (the title bar steps aside while you type). Capitals are automatic (calls and parks in capitals; names and comments capitalized like normal writing).
2. **Tab** on the keyboard moves Call → Sent → Rcvd → P2P → Comment (LOG or Enter logs) with the text highlighted. The green **LOG** key logs from any box.
3. The page shrinks above the keyboard and scrolls so the form stays in view (on a small phone held sideways, the box you're typing in). **Drag the page** (or tap ▼ or Back) and the keyboard drops; **tap a box** to bring it back.
4. **abc** switches that one box to the phone's own keyboard (handy for long comments). Turn the ham keyboard off in Settings → My station if you prefer the phone's keyboard everywhere.
5. Plug in or pair a **Bluetooth or USB keyboard** and the ham keyboard steps aside by itself; Tab and Enter then work the same as on the computer. (The Windows/Linux program never shows it - it uses your real keyboard.)

Radio over USB-OTG

1. In Settings → Radio, pick your radio (the same list as Windows - Yaesu, Icom and Xiegu; Flrig and Hamlib are Windows-only).
2. Plug the radio's USB cable into a **USB-OTG adapter** on the tablet and turn the radio on.
3. Tap **Connect radio** and allow USB access when Android asks.
4. Radios with two USB ports (FT-710, FT-991A, IC-705): use **USB port 1**. Match the speed to the radio's CAT menu.

Sending from the tablet

Send to QRZ needs only internet. **Send to LOC** to ACLog, DXKeeper, Log4OM or HRD needs the tablet on the **same Wi-Fi as the home PC** - use the PC's address (like 192.168.1.20) instead of 127.0.0.1, and allow the port through Windows Firewall. Wavelog/Cloudlog works from anywhere online. Contacts wait (and keep their count on the button) until you're back on Wi-Fi.

Part 4 - Linux

The same program runs on Linux - Ubuntu, Linux Mint, Debian, Raspberry Pi OS, Fedora, Arch and others with a desktop. Everything works the same as on Windows.

Installing

1. Unzip **Raw_Ham_Lite_Logger_Linux.zip** and open a terminal in its **Raw Ham Lite Logger Linux** folder (right-click inside the folder → **Open in Terminal**).
2. Type the line below and press Enter. Answer **Y** to its questions; it may ask for your password.

```
bash install-linux.sh
```

3. It installs what the logger needs (**python3-tk**, **python3-serial** and the **secret-tool** keyring helper), copies the program to **~/.local/share/rawham-lite-logger**, and adds a menu entry and desktop icon with the logo.
4. If it adds you to the **dialout** group (needed for USB radios), **log out and back in** once.
5. Start it from your app menu or the desktop icon. To update later, unzip the new version and run **bash install-linux.sh** again - your log and settings are kept.

Linux differences

Thing	On Linux
Radio port	Looks like **/dev/ttyUSB0** or **/dev/ttyACM0** instead of COM3. FT-710 / FT-991A: the Enhanced port is usually **ttyUSB0**. IC-705: **ttyACM0**.
"Can't open /dev/ttyUSB0"	You're not in the dialout group yet: run **sudo usermod -aG dialout $USER**, then log out and back in.
Flrig and Hamlib	Both are native on Linux (**sudo apt install flrig** or **libhamlib-utils**). Pick them in Settings → Radio for any radio they support.
Passwords	Kept in your desktop keyring (GNOME Keyring / KDE Wallet). Without a keyring they stay in a settings file only your user can read.
Log of choice	ACLog, DXKeeper, Log4OM and HRD are Windows programs - send to them over your home network using that PC's address. Wavelog/Cloudlog and WSJT-X work the same.
Exported ADIF files	**~/Documents/Raw Ham Logger ADIF**.

Part 5 - Updating, sharing and help

Updating to a newer version

Your log, settings and saved lists are kept by every update. When a new version is out, the version number in the logger turns **red**. On Windows, click it and the logger downloads and runs the new Setup for you. On Android, tap it to download the new APK. Or get any version from the download page.

1. **Windows:** run the new **Raw Ham Lite Logger Setup** - it replaces the old program. (Close the logger first.)
2. **Android:** install the new **RawHamLiteLogger.apk** over the old one.
3. **Linux:** unzip the new **Raw_Ham_Lite_Logger_Linux.zip** and run **bash install-linux.sh** again.
4. **Python file:** replace **rawhams_lite_logger.py** with the new one.

Sharing with a friend

Send them the download page, **rawhamradio.github.io/rawham-lite-logger** - it has all three versions and this guide. Never share **rawhams_lite_data.json** or **rawhams_lite.db**. Tell them:

1. Enter **their own** call, grid and QRZ login.
2. Pick their radio in Settings → Radio. Not in the list? Use Flrig or Hamlib on Windows/Linux.
3. No QRZ subscription? Pick their home logger in Settings → Log of choice instead.

Questions or found a bug?

Email me at **KJ5GVQ+logger@gmail.com**. Tell me what you were doing, what you expected, and what happened instead. Say whether it's Windows, Android or Linux, and the version number. A screenshot helps a lot. I read every one.

Keeping your log and passwords safe

1. Your QRZ password and API keys are **encrypted** - on Windows with your Windows login, on Android with the phone's secure key store - and Android leaves them out of Google backups.
2. Only WSJT-X on your own PC can add contacts, so nobody on campground or club Wi-Fi can slip in fake QSOs.
3. QRZ, spotting and downloads all use encrypted (https) connections. If you enter a Wavelog address that starts with http:// on the internet, the logger warns you.
4. Only share the program files or the APK - never rawhams_lite_data.json or rawhams_lite.db.

Your files

File / folder	What it is
The program (Windows)	Installed in your user folder: AppData\Local\Programs\Raw Ham Lite Logger. Remove it with Settings → Apps → The Raw Ham's Lite Logger.
Your log and settings (Windows)	AppData\Local\Raw Ham Lite Logger - kept when you update or uninstall.
Python / Linux versions	Your log and settings sit next to rawhams_lite_logger.py.
rawhams_lite_data.json	Your settings and current activation. Passwords and keys in it are encrypted and only open for your Windows login on this PC. Still private.
rawhams_lite.db	Worked-before history, saved lookups, park lists, offline callbook.
Documents\Raw Ham Logger ADIF	Your exported ADIF files.

Appendix - running the Python file (instead of Setup)

Only for Linux (the installer does this for you) or if you prefer Python on Windows. The Windows Setup above needs none of this.

1. Go to **python.org/downloads**, click **Download Python** and run it. On the first screen tick **"Add python.exe to PATH"**, then **Install Now**. (Missed it? Run the installer again, choose **Modify**, and tick **Add Python to environment variables**.)
2. Press the Windows key, type **cmd**, press Enter, and type:

```
py -m pip install pyserial access-parser
```

3. Put **rawhams_lite_logger.py**, **logo.ico** and **logo.png** (from the Linux folder) together in a folder, e.g. Documents\Raw Ham Logger.
4. Double-click **rawhams_lite_logger.py** (choose **Python** if asked). It offers to make a desktop shortcut with the logo - click **Yes**. (Or later: Settings → Make desktop shortcut.)

Troubleshooting

Problem	Fix
"Windows protected your PC"	Click **More info** → **Run anyway**. Windows says this for any new program from a small publisher.
Antivirus removed or blocked the program	Allow / restore **Raw Ham Lite Logger** in your antivirus, then run Setup again. It's a false alarm.
Python file: 'py' is not recognized	Python wasn't added to PATH - see the Appendix.

Problem	Fix
Raw Roster stays empty	Settings → WSJT-X → **Listen to WSJT-X** must be ticked, and the port must match WSJT-X's UDP Server port (usually 2237).
LoTW: "couldn't find that station location"	Pick the location from the list in Settings → QRZ - it must match TQSL exactly.
Photo or Share says it needs Pillow (Linux)	Run **bash install-linux.sh** again and answer Y to Pillow, or: sudo apt install python3-pil
Send buttons are greyed out	A special event call is set (Settings → My station). Clear it, or use Export ADIF for the event log.
Python file: radio control is off	Run the pip line in the Appendix again.
Radio didn't answer	Check the radio type in Settings → Radio, the COM port (Yaesu: the **Enhanced** one), and that the speed matches the radio's CAT menu. Try toggling **DTR/RTS off**. Close other programs using the radio.
"Can't reach Flrig / rigctld"	Start Flrig or rigctld first and check the port (12345 / 4532).
"Saved password couldn't be unlocked"	Passwords are locked to the Windows user and PC they were saved on. Type them again in Settings.
WSJT-X on another PC is ignored	Tick Also accept contacts from other computers in Settings → WSJT-X.
Worked-before seems out of date	At home, click Settings → Update everything now. The line in Offline data shows how many came from QRZ and your log, and when.
Toolbar or window too big on a small screen	Drag the window edges smaller - it's built to fit a laptop screen.
Couldn't read the ACLog file	Export an ADIF from ACLog and use Settings → Offline data → Import worked-before from an ADIF file.
Send to ACLog / log of choice fails	Click Test connection in Settings → Log of choice. The logger must be open with its API/network option on (see Step 4). From a tablet, use the PC's address and allow the port in Windows Firewall.
No park names / no drop-down list	Connect to the internet and click Settings → Offline data → Update park & summit lists.
Names don't fill in	Run Download offline callbook (US/Canada), or set up QRZ for DX calls.
QRZ test fails	Use your QRZ **password**, not the Logbook API key, and check your XML subscription.
Spot FAILED	Check your internet. SOTA and LLOTA always open a web page instead.
A download says FAILED	Try again later - the website may be busy. Note the message for help.

Problem	Fix
WSJT-X contacts don't appear	Tick Listen to WSJT-X (Settings → WSJT-X) and enter your park first. In WSJT-X: Settings → Reporting → UDP Server 127.0.0.1 port 2237, and tick Accept UDP requests. If another program holds the port, use a multicast address in both.
Send to QRZ fails	Use the Logbook **API key** (not your password) in Settings → QRZ. QRZ needs a subscription for logbook uploads and downloads.
Can't find exported files	Documents\Raw Ham Logger ADIF, or see Settings → Offline data → Choose ADIF export folder.

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